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Electric Vehicle  
Codec Technologies India  
Internship 2026

## Off-Grid Solar EV Charging Station

HOMER-style hourly dispatch and sizing for Ghaziabad on real NASA POWER 2025 irradiance

### AT A GLANCE

**1,676 kWh/m<sup>2</sup>**

Annual GHI (NASA POWER 2025)

**1,423 kWh/kWp**

Specific yield, 28 deg tilt

**55,672 kWh/yr**

EV demand (synthetic)

**70 kWp / 125 kWh**

Chosen design

**3.86 %**

LPSP

**44.2 %**

Curtailed PV

**Rs 17.20 / 12.24 per kWh**

LCOE off-grid vs grid

**11 / 11**

Tests re-run here

### VERDICT

TESTS 11/11

RE-RUN HERE

REAL DATA (solar)

SYNTHETIC LOAD

DECLARED SUBSTITUTION

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## 1. Objective and brief mapping

Brief: design an off-grid solar EV charging station, simulate it in HOMER or PVsyst, include battery storage, and calculate daily solar input and EV charging demand. HOMER Pro and PVsyst were not available to me, so I wrote a **HOMER-style techno-economic simulation in Python**: real hourly irradiance -> PV model -> synthetic EV load -> hourly PV/battery/unmet dispatch over 8,760 h -> 13 x 19 sizing grid -> NPC/LCOE, plus an Excel workbook with the economics as live formulas. The method mirrors HOMER's definitions (load-following dispatch, CRF, NPC, LCOE); the software is my own.

**Data provenance:** irradiance and air temperature are **real** - 8,760 hourly values (ALLSKY\_SFC\_SW\_DWN, T2M) for 28.67 N, 77.42 E, calendar 2025, downloaded from the NASA POWER API and saved unmodified in data/nasa\_power\_hourly\_2025.csv. The EV demand is **synthetic** (seeded stochastic arrivals). Costs are cited (MNRE, UPERC, Ember) or declared as assumptions.

## 2. Model and architecture

Solar: declination, hour angle (IST meridian 82.5 E), Erbs diffuse fraction, isotropic-sky transposition to a 28 deg south-facing plane, cell temperature  $T_c = T_{amb} + (NOCT - 20)/800 POA$ ,  $P_{dc} = kWp POA/1000 [1 + \gamma (T_c - 25)] f_{soil} f_{wire}$ , inverter 96 % with DC:AC 1.15 clipping. Battery: LiFePO4 energy-in-tank, SOC 10-90 %, 92 % round-trip split  $\sqrt{\eta}$  per leg, 0.5C limits. Load: 2 x 7.4 kW AC + 1 x 30 kW DC chargers, Poisson daily arrivals of scooters / AC cars / DC cars, first-come-first-served port queue with 60 min abandonment.

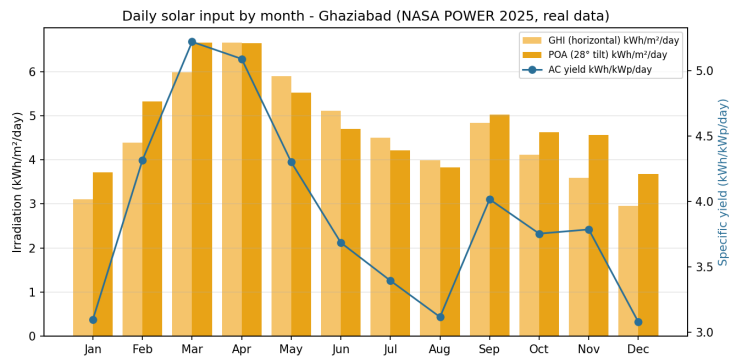


Fig. 1 - Real NASA POWER 2025 daily GHI and modelled POA / AC yield by month for Ghaziabad; the monsoon (Jul-Aug) and winter-fog (Dec-Jan) dips are both visible.

## 3. Method and implementation

- Dispatch:  $charge\_t = \min(surplus, (E_{max} - E_t)/\sqrt{\eta}, 0.5C)$ ;  $discharge\_t = \min(deficit, (E_t - E_{min}) \sqrt{\eta}, 0.5C \sqrt{\eta})$ ; unmet and curtailed tracked hourly; LPSP =  $\sum unmet / \sum load$ .
- Sizing: 13 PV sizes (30-150 kWp) x 19 battery sizes (50-500 kWh) = 247 dispatch runs; the design is the minimum-NPC combination with LPSP  $\leq 5\%$ .
- Economics (20 yr, 8 % real): CRF = 0.1019; NPC =  $capex + O\&M/CRF + PV(inverter + battery replacement \text{ at } yr \ 10)$ ; LCOE = NPC / discounted served kWh; grid alternative at the UPERC LMV-11 EV tariff Rs 7.46/kWh escalating 3 %/yr.
- Workbook results/solar\_ev\_station.xlsx: 29 live economics formulas, MINIFS/INDEX/MATCH design pick, recalculated headlessly in LibreOffice with 0 error cells; formula LCOE = Python LCOE within 3e-5 Rs/kWh.

## 4. Results (results/summary.json, monthly\_energy\_balance.csv)

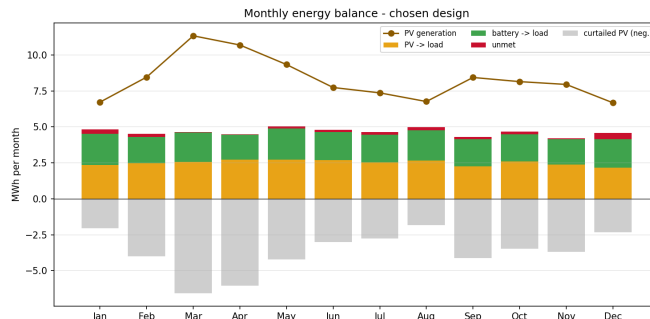


Fig. 2 - Monthly energy balance (MWh): PV 6.7-11.3, load 4.2-5.0, unmet concentrated in Dec (0.44) and Jan (0.31), curtailment peaking in Mar-Apr (6.6 / 6.0).

| Quantity                      | Value                                                   | Quantity               | Value                       |
|-------------------------------|---------------------------------------------------------|------------------------|-----------------------------|
| Annual GHI / mean daily       | 1,676 kWh/m <sup>2</sup> / 4.59 kWh/m <sup>2</sup> /day | PV generation (70 kWp) | 99,602 kWh/yr               |
| Specific AC yield             | 1,423 kWh/kWp/yr                                        | Load served / unmet    | 53,524 / 2,148 kWh          |
| Annual EV demand / mean daily | 55,672 kWh / 152.5 kWh                                  | LPSP                   | 3.86 % (target $\leq 5\%$ ) |

**KEY TAKEAWAY** On real 2025 Ghaziabad irradiance a 70 kWp / 125 kWh off-grid station serves 96 % of a 55.7 MWh/yr demand at Rs 17.20/kWh - 40 % dearer than grid charging because 44 % of the PV must be curtailed.

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| Quantity                             | Value                       | Quantity                          | Value                             |
|--------------------------------------|-----------------------------|-----------------------------------|-----------------------------------|
| Peak hourly load                     | 44.4 kW (44.8 kW installed) | PV direct / via battery           | 30,086 / 23,438 kWh               |
| Sessions served / lost to queue      | 5,257 / 848 of 6,105        | Curtailed PV                      | 44,053 kWh (44.2 %)               |
| Energy by class 2W / AC car / DC car | 4,890 / 11,649 / 39,137 kWh | Battery cycles                    | 187.5 EFC/yr, 15.7 h of mean load |
| Feasible designs                     | 178 of 247                  | Capex / NPC                       | Rs 71.0 L / Rs 90.38 L            |
| LCOE off-grid / grid alternative     | Rs 17.20 / 12.24 per kWh    | Payback vs grid / on Rs 15 retail | 18.3 / 10.0 yr                    |

### 4b. Sizing-grid neighbourhood of the chosen design (results/sizing\_grid.csv)

The design sits on the LPSP = 5 % frontier: one battery step smaller (100 kWh) fails the target, one PV step smaller (60 kWp) needs 150 kWh of battery and costs Rs 1.05 L more in NPC. Every step up in PV raises curtailment faster than it lowers unmet energy, which is why the LCOE minimum is at the smallest feasible array.

| PV kWp    | Battery kWh | LPSP          | Served kWh    | Curtailed     | Cycles/yr    | Capex Rs L  | NPC Rs L    | LCOE Rs/kWh  | Feasible                |
|-----------|-------------|---------------|---------------|---------------|--------------|-------------|-------------|--------------|-------------------------|
| 60        | 125         | 5.29 %        | 52,729        | 35.9 %        | 189.8        | 66.2        | 84.3        | 16.29        | no                      |
| 60        | 150         | 3.50 %        | 53,724        | 34.6 %        | 164.8        | 71.2        | 91.4        | 17.33        | yes (2nd cheapest)      |
| 70        | 100         | 7.09 %        | 51,723        | 46.2 %        | 216.4        | 66.0        | 83.3        | 16.40        | no                      |
| <b>70</b> | <b>125</b>  | <b>3.86 %</b> | <b>53,524</b> | <b>44.2 %</b> | <b>187.5</b> | <b>71.0</b> | <b>90.4</b> | <b>17.20</b> | <b>chosen (min NPC)</b> |
| 70        | 150         | 2.16 %        | 54,470        | 43.2 %        | 162.6        | 76.0        | 97.5        | 18.23        | yes                     |
| 80        | 125         | 2.97 %        | 54,019        | 50.8 %        | 185.2        | 75.8        | 96.4        | 18.18        | yes (3rd)               |
| 90        | 125         | 2.34 %        | 54,372        | 56.0 %        | 183.2        | 80.6        | 102.5       | 19.20        | yes                     |

### 5. Verification

Re-run here with the fixture-aware runner (pytest not installed): **11 passed, 0 failed** in 3.5 s.

| Test                                                                   | Checks                                                                           | Result  |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------|
| test_nasa_file_complete                                                | 8,760 rows, no fill values, plausible annual GHI                                 | PASS    |
| test_poa_and_yield_plausible                                           | POA >= 0 and zero at night, yield within 1,200-1,800 kWh/kWp, clipping respected | PASS    |
| test_load_reproducible_and_positive                                    | seeded load identical between runs, >= 0, never above installed charger power    | PASS    |
| test_hourly_energy_balance_closes / test_battery_energy_balance_closes | hourly balance error 0.0 kWh; battery balance error 2e-12 kWh                    | PASS x2 |
| test_soc_within_bounds / test_c_rate_limits                            | SOC stays in 10-90 %, 0.5C charge and discharge limits respected                 | PASS x2 |
| test_no_battery_means_more_unmet                                       | removing the battery raises LPSP                                                 | PASS    |
| test_chosen_design_meets_lpssp                                         | 70 kWp / 125 kWh meets LPSP <= 5 % and is the min-NPC feasible point             | PASS    |
| test_economics_positive_and_consistent                                 | LCOE > 0, NPC > capex, CRF(8 %, 20 yr) = 0.1019                                  | PASS    |
| test_summary_json_exists_and_consistent                                | served + unmet = load in summary.json                                            | PASS    |

### 6. Data integrity and declared substitutions

- **HOMER Pro / PVSyst not used** - README: "HOMER Pro / PVSyst were not available to me, so I wrote a HOMER-style techno-economic simulation in Python (hourly dispatch, sizing search, NPC/LCOE). The method mirrors HOMER's; the software is my own." My dispatch is load-following only; HOMER would also try cycle-charging, a genset or grid export.
- Solar and temperature are real NASA POWER data (0.5 x 0.625 deg grid cell, +/-5-10 % POA error normal); EV demand is synthetic and the sizing would change with measured session logs. Single weather year repeated for 20 years, no PV degradation, hourly resolution.
- Prices: MNRE benchmark Rs 48,000/kWp is FY2019-20; battery Rs 20,000/kWh is my estimate anchored on Ember's USD 125/kWh utility figure; chargers, civil, O&M, escalation and retail price are declared assumptions in the workbook's Inputs sheet.

### 7. Learnings

The honest conclusion is that a fully off-grid station costs about 40 % more per delivered kWh than grid charging at the UP EV tariff. The driver is winter: to hold LPSP under 5 % in December-January fog the array must be oversized, so 4 of every 9 kWh produced in March-April are thrown away. The 848 lost sessions are a port problem, not an energy problem - two AC ports are the bottleneck and a third AC port is the first upgrade I would test. A grid-tied hybrid that exports the surplus is the obvious next study, and it is one cell edit away in the workbook.

**KEY TAKEAWAY** On real 2025 Ghaziabad irradiance a 70 kWp / 125 kWh off-grid station serves 96 % of a 55.7 MWh/yr demand at Rs 17.20/kWh - 40 % dearer than grid charging because 44 % of the PV must be curtailed.